WORKED SOLUTION

STEP I 1988, Question 8

Pure • Geometry in Three Dimensions

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Read the question before you go on. It is in the STEP Support Programme's question database:

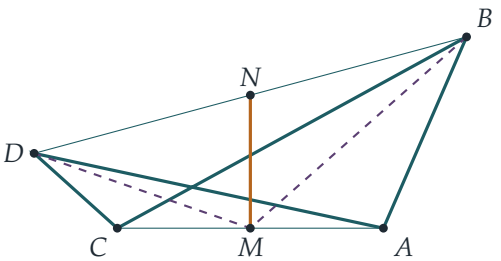
step.maths.org/questions/88-s1-q8

The key idea

Each diagonal splits the quadrilateral into two triangles, and the equal opposite sides make those two triangles congruent (SSS). Congruent triangles have equal medians, so the midpoint of one diagonal is equidistant from the two ends of the other. A point equidistant from both ends of a segment, joined to the segment's midpoint, gives a line perpendicular to that segment: this is just the isosceles triangle.

Solution

Let M be the midpoint of AC and N the midpoint of BD.



The figure is a projection of a skew quadrilateral; the dashed lines MB and MD are the medians used below, and the amber segment is MN.

First, $M \neq N$

The line MN only makes sense if M and N are different points. If $M = N$, the diagonals AC and BD would bisect each other at a common point. Two lines through a common point lie in a plane, so A, B, C, D would all lie in that plane, and $ABCD$ would be a (planar) parallelogram. Since $ABCD$ is skew, $M \neq N$. This is exactly where the word "skew" is used: for a parallelogram the opposite sides are equal but the two midpoints

coincide.

MN is perpendicular to BD

Look at the triangles ABC and CDA, which share the diagonal AC. We have

$$AB = CD, \qquad BC = DA, \qquad CA = AC,$$

so the triangles are congruent (SSS), with $A \leftrightarrow C$, $B \leftrightarrow D$, $C \leftrightarrow A$. Under this correspondence the median from B to the midpoint of AC in triangle ABC matches the median from D to the midpoint of CA in triangle CDA. Both medians end at the same point M, so

$$MB = MD.$$

Ask yourself Why must congruent triangles have equal medians?

A median is determined entirely by the side lengths. For instance, by the cosine rule (or Apollonius's theorem), the median m from B in triangle ABC satisfies $4m^2 = 2AB^2 + 2BC^2 - AC^2$. The same side lengths give the same median, so $MB^2 = \frac{1}{4}(2AB^2 + 2BC^2 - AC^2) = \frac{1}{4}(2CD^2 + 2DA^2 - CA^2) = MD^2$.

Now M is equidistant from B and D, and N is the midpoint of BD. The triangles MNB and MND have $MB = MD$, $NB = ND$ and the common side MN, so they are congruent (SSS), and the angles $\angle MNB$ and $\angle MND$ are equal. They are also supplementary, because B, N, D lie on a line. So each is a right angle: $MN \perp BD$. (The triangles are genuine, not degenerate: if M lay on the line BD with $MB = MD$, then M would be the midpoint N, which we have ruled out.)

MN is perpendicular to AC

The same argument works with the roles of the diagonals swapped. Triangles ABD and CDB share the diagonal BD, and

$$AB = CD, \qquad AD = CB, \qquad BD = DB,$$

so they are congruent, with $A \leftrightarrow C$, $B \leftrightarrow D$, $D \leftrightarrow B$. The medians from A and from C both end at the midpoint N of BD, so $NA = NC$. Since M is the midpoint of AC and $N \neq M$, the isosceles-triangle argument above gives $MN \perp AC$.

So the line joining the midpoints of the diagonals is perpendicular to each diagonal, as required.

Or with vectors

Take the origin at M, so that $C = -A$ as position vectors, and $\overrightarrow{MN} = \frac{1}{2}(\mathbf{b} + \mathbf{d})$. Squaring the two given equalities,

$$AB^2 = CD^2 \iff |\mathbf{b}|^2 - 2\mathbf{a}\cdot\mathbf{b} = |\mathbf{d}|^2 + 2\mathbf{a}\cdot\mathbf{d},$$
$$BC^2 = AD^2 \iff |\mathbf{b}|^2 + 2\mathbf{a}\cdot\mathbf{b} = |\mathbf{d}|^2 - 2\mathbf{a}\cdot\mathbf{d}.$$

Subtracting gives $4\mathbf{a}\cdot\mathbf{b} = -4\mathbf{a}\cdot\mathbf{d}$, that is $\mathbf{a}\cdot(\mathbf{b}+\mathbf{d}) = 0$. Since $\overrightarrow{AC} = -2\mathbf{a}$, this says $\overrightarrow{MN} \perp AC$. Putting the origin at N instead gives $MN \perp BD$ in the same way.

Checks

Try $A = (1,0,0)$, $B = (1,2,1)$, $C = (-1,0,0)$, $D = (-1,-2,1)$. Then $AB = CD = \sqrt{5}$ and $BC = AD = 3$, and the four points are not coplanar (the scalar triple product $\overrightarrow{AB}\cdot(\overrightarrow{AC}\times\overrightarrow{AD})$ is 8). Here $M = (0,0,0)$ and $N = (0,0,1)$, so MN is along the z-axis, while AC is along the x-axis and $\overrightarrow{BD} = (-2,-4,0)$ has no z-component: both are perpendicular to MN. We also tested 20 000 random skew quadrilaterals with equal opposite sides; the cosine of the angle between MN and either diagonal never exceeded 4×10^{-12}.

Common slips

- Not explaining why $M \neq N$. Without the skew condition the result fails, since for a parallelogram there is no line MN at all.
- Claiming $MB = MD$ "by symmetry" without naming the congruent triangles.
- Proving perpendicularity to one diagonal only. The second needs a different pair of congruent triangles.

Beyond STEP

In the example, the half-turn about the z-axis (the line MN) swaps A with C and B with D. This is true in general: every skew quadrilateral with equal opposite sides is carried onto itself by a rotation through 180° about MN. (The rotation fixes M and reverses every direction perpendicular to MN, so it sends A to C; likewise it sends B to D.) Conversely, the half-turn makes the equal sides obvious. Seeing geometric facts as consequences of a symmetry, an isometry of space, is a central idea in first-year geometry and group theory; Allenby's *Rings, Fields and Groups* develops the group side, and Siklos's *Advanced Problems in Mathematics* has more STEP-style geometry.

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